Zolotarev 1a

A.C. Antoulas[1], I-V. Gosea[2], and C. Poussot-Vassal*[3]

[1]Department of Electrical and Computer Engineering, Rice University, Houston (aca@rice.edu)
[2]Max Planck Institute, CSC Group, Magdeburg, Germany (gosea@mpi-magdeburg.mpg.de)
[3]ONERA, DTIS, Université de Toulouse, France (charles.poussot-vassal@onera.fr)

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Abstract

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*Corresponding author: charles.poussot-vassal@onera.fr

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2 $r = 3$

2.1 Numerator, denominator, poles and zeros

	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00044$		AAA-sign-L1000, $\sigma_3 = 0.0005$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	-0.011	$6.5 \cdot 10^{-4}$	-70.0	12.0
z^2	2.6	0	3.7	$1.8 \cdot 10^{-16}$	2.6	$-2.2 \cdot 10^{-3}$	0.95	0.064
z	0	0	$-6.8 \cdot 10^{-15}$	$-1.2 \cdot 10^{-15}$	$9.9 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	-150.0	27.0
1.0	0.65	0	1.5	$1.9 \cdot 10^{-16}$	0.65	$-6.1 \cdot 10^{-4}$	-0.36	0.082
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-4.2 \cdot 10^{-15}$	$-1.3 \cdot 10^{-15}$	$-6.6 \cdot 10^{-3}$	$2.1 \cdot 10^{-3}$	-180.0	31.0
z	2.2	0	4.2	$4.4 \cdot 10^{-16}$	2.3	$-1.9 \cdot 10^{-3}$	-0.38	0.084
1.0	0	0	$-2.8 \cdot 10^{-15}$	$-2.0 \cdot 10^{-16}$	$7.3 \cdot 10^{-3}$	$2.0 \cdot 10^{-4}$	-44.0	7.6

Table 1: Case `1a` (cpv macos), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00037$		AAA-sign-L1000, $\sigma_3 = 0.00037$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	$-2.4 \cdot 10^{-11}$	$2.7 \cdot 10^{-13}$	$-9.5 \cdot 10^{-15}$	$5.6 \cdot 10^{-13}$
z^2	2.6	0	3.7	$-1.8 \cdot 10^{-16}$	2.6	$-5.8 \cdot 10^{-15}$	2.6	$-4.2 \cdot 10^{-15}$
z	0	0	$-5.1 \cdot 10^{-15}$	$6.8 \cdot 10^{-16}$	$-5.4 \cdot 10^{-11}$	$6.3 \cdot 10^{-13}$	$-2.1 \cdot 10^{-14}$	$1.2 \cdot 10^{-12}$
1.0	0.65	0	1.5	$-2.4 \cdot 10^{-16}$	0.65	$-9.7 \cdot 10^{-15}$	0.65	$-6.4 \cdot 10^{-15}$
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-3.2 \cdot 10^{-15}$	$3.0 \cdot 10^{-18}$	$-6.2 \cdot 10^{-11}$	$7.1 \cdot 10^{-13}$	$-2.3 \cdot 10^{-14}$	$1.4 \cdot 10^{-12}$
z	2.2	0	4.2	$-4.0 \cdot 10^{-16}$	2.2	$-9.9 \cdot 10^{-15}$	2.2	$-9.8 \cdot 10^{-15}$
1.0	0	0	$-1.8 \cdot 10^{-15}$	$4.3 \cdot 10^{-16}$	$-1.6 \cdot 10^{-11}$	$1.9 \cdot 10^{-13}$	$-7.1 \cdot 10^{-15}$	$3.6 \cdot 10^{-13}$

Table 2: Case `1a` (cpv winos), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	0	$6.58 \cdot 10^{-16} + 4.81 \cdot 10^{-17}\imath$	$-0.00323 - 9.0 \cdot 10^{-5}\imath$	$-0.00181 + 0.493\imath$
2.0	$1.5\imath$	$2.14 \cdot 10^{-15} - 2.06\imath$	$0.00557 + 1.5\imath$	$-0.00165 - 0.494\imath$
3.0	$-1.5\imath$	$1.49 \cdot 10^{-15} + 2.06\imath$	$0.00428 - 1.5\imath$	$180.0 - 31.3\imath$

Table 3: Case `1a` (cpv macos), $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	0	$4.35 \cdot 10^{-16} - 1.02 \cdot 10^{-16}\imath$	$6.9 \cdot 10^{-12} - 8.47 \cdot 10^{-14}\imath$	$3.42 \cdot 10^{-15} - 1.59 \cdot 10^{-13}\imath$
2.0	$1.5\imath$	$1.25 \cdot 10^{-15} - 2.06\imath$	$2.76 \cdot 10^{-11} + 1.5\imath$	$1.22 \cdot 10^{-14} + 1.5\imath$
3.0	$-1.5\imath$	$1.47 \cdot 10^{-15} + 2.06\imath$	$2.76 \cdot 10^{-11} - 1.5\imath$	$7.73 \cdot 10^{-15} - 1.5\imath$

Table 4: Case `1a` (cpv winos), $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	NaN	NaN	$-0.00241 + 0.5\imath$	$-0.00236 + 1.2 \cdot 10^{-4}\imath$
2.0	$0.5\imath$	$9.48 \cdot 10^{-16} - 0.637\imath$	$-0.00245 - 0.501\imath$	$0.00667 - 1.48\imath$
3.0	$-0.5\imath$	$8.89 \cdot 10^{-16} + 0.637\imath$	$237.0 + 13.9\imath$	$0.00871 + 1.48\imath$

Table 5: Case `1a` (cpv macos), $r = 3$, Z4: zeros.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	NaN	NaN	$9.21 \cdot 10^{-12} + 0.5\imath$	$4.98 \cdot 10^{-15} + 0.5\imath$
2.0	$0.5\imath$	$7.16 \cdot 10^{-16} + 0.637\imath$	$9.2 \cdot 10^{-12} - 0.5\imath$	$2.0 \cdot 10^{-15} - 0.5\imath$
3.0	$-0.5\imath$	$6.45 \cdot 10^{-16} - 0.637\imath$	$1.09 \cdot 10^{11} + 1.22 \cdot 10^{9}\imath$	$7.82 \cdot 10^{10} + 4.66 \cdot 10^{12}\imath$

Table 6: Case `1a` (cpv winos), $r = 3$, Z4: zeros.

2.2 Plots

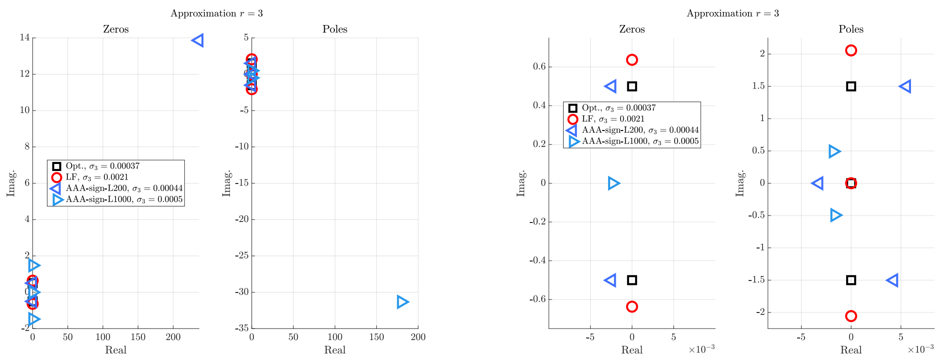

Figure 1: Case `'1a'`, $r = 3$. Left: zeros (left) and poles (right). Right: discarding the large values.

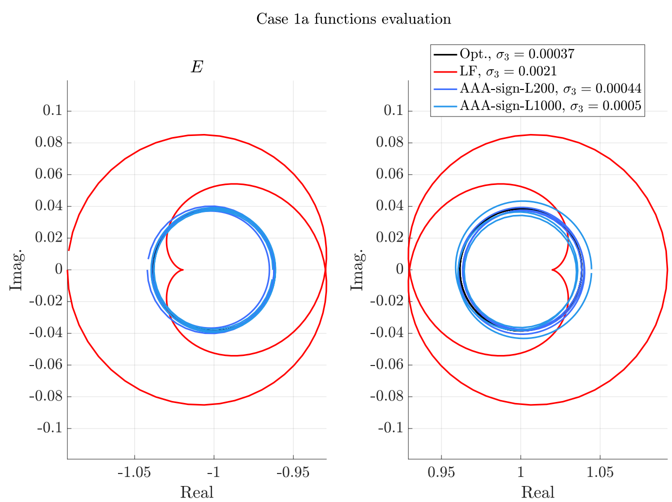

Figure 2: Case `'1a'`, $r = 3$. E and F evaluations.

3 $r = 4$

3.1 Numerator, denominator, poles and zeros

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	-140.0	16.0	-150.0	7.3
z^3	3.5	0	3.7	$4.1 \cdot 10^{-16}$	4.7	-0.29	4.8	-0.22
z^2	0	0	$-4.6 \cdot 10^{-15}$	$3.8 \cdot 10^{-16}$	-650.0	70.0	-660.0	33.0
z	2.6	0	3.2	$2.0 \cdot 10^{-16}$	5.5	-0.52	5.6	-0.34
1.0	0	0	$-1.3 \cdot 10^{-15}$	$-1.6 \cdot 10^{-15}$	-80.0	8.8	-82.0	4.1
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-2.7 \cdot 10^{-15}$	$-8.4 \cdot 10^{-16}$	-500.0	54.0	-510.0	25.0
z^2	4.5	0	5.2	$8.2 \cdot 10^{-16}$	7.9	-0.7	7.9	-0.49
z	0	0	$-3.5 \cdot 10^{-15}$	$-1.1 \cdot 10^{-15}$	-370.0	41.0	-380.0	19.0
1.0	0.56	0	0.75	$1.0 \cdot 10^{-16}$	1.4	-0.13	1.4	-0.078

Table 7: Case `1a` (cpv macos), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	$-5.4 \cdot 10^{-3}$	$-3.1 \cdot 10^{-3}$	$-4.8 \cdot 10^{-3}$	$-2.8 \cdot 10^{-3}$
z^3	3.5	0	3.7	$1.8 \cdot 10^{-16}$	3.5	$-3.7 \cdot 10^{-3}$	3.5	$-3.8 \cdot 10^{-3}$
z^2	0	0	$-2.8 \cdot 10^{-15}$	$-9.9 \cdot 10^{-16}$	-0.07	$-2.6 \cdot 10^{-3}$	-0.078	$-2.5 \cdot 10^{-3}$
z	2.6	0	3.2	$-5.5 \cdot 10^{-16}$	2.6	-0.011	2.6	-0.011
1.0	0	0	$-1.8 \cdot 10^{-15}$	$-7.7 \cdot 10^{-16}$	-0.015	$1.3 \cdot 10^{-3}$	-0.017	$1.4 \cdot 10^{-3}$
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-4.6 \cdot 10^{-16}$	$-1.2 \cdot 10^{-15}$	-0.036	$-6.9 \cdot 10^{-3}$	-0.038	$-6.5 \cdot 10^{-3}$
z^2	4.5	0	5.2	$2.7 \cdot 10^{-16}$	4.5	-0.011	4.5	-0.011
z	0	0	$-4.2 \cdot 10^{-15}$	$-4.2 \cdot 10^{-16}$	-0.054	$2.2 \cdot 10^{-3}$	-0.062	$2.5 \cdot 10^{-3}$
1.0	0.56	0	0.75	$-5.2 \cdot 10^{-16}$	0.57	$-3.3 \cdot 10^{-3}$	0.57	$-3.3 \cdot 10^{-3}$

Table 8: Case `1a` (cpv winos), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09\imath$	$3.21 \cdot 10^{-16} - 0.384\imath$	$0.00377 + 6.65 \cdot 10^{-5}\imath$	$0.00375 - 2.07 \cdot 10^{-5}\imath$
2.0	$-1.94 \cdot 10^{-16} - 2.09\imath$	$3.09 \cdot 10^{-16} + 0.384\imath$	$0.00535 - 0.864\imath$	$0.00513 + 0.864\imath$
3.0	$-1.39 \cdot 10^{-17} + 0.359\imath$	$1.12 \cdot 10^{-15} - 2.25\imath$	$0.00516 + 0.864\imath$	$0.00532 - 0.864\imath$
4.0	$-1.39 \cdot 10^{-17} - 0.359\imath$	$9.09 \cdot 10^{-16} + 2.25\imath$	$499.0 - 54.4\imath$	$507.0 - 25.2\imath$

Table 9: Case `1a` (cpv macos), $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09\imath$	$5.75 \cdot 10^{-16} + 0.384\imath$	$0.00636 + 0.36\imath$	$0.00731 + 0.36\imath$
2.0	$-1.94 \cdot 10^{-16} - 2.09\imath$	$2.75 \cdot 10^{-16} - 0.384\imath$	$0.0052 - 0.361\imath$	$0.00612 - 0.361\imath$
3.0	$-1.39 \cdot 10^{-17} + 0.359\imath$	$-1.15 \cdot 10^{-16} - 2.25\imath$	$0.00935 - 2.1\imath$	$0.00961 - 2.09\imath$
4.0	$-1.39 \cdot 10^{-17} - 0.359\imath$	$-2.71 \cdot 10^{-16} + 2.25\imath$	$0.0147 + 2.1\imath$	$0.0149 + 2.1\imath$

Table 10: Case `1a` (cpv winos), $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00403 - 0.358\imath$	$0.00404 + 0.358\imath$
2.0	0	$4.1 \cdot 10^{-16} + 5.92 \cdot 10^{-16}\imath$	$0.00406 + 0.358\imath$	$0.00402 - 0.358\imath$
3.0	$0.866\imath$	$3.69 \cdot 10^{-16} + 0.929\imath$	$0.0128 - 2.08\imath$	$0.0127 - 2.09\imath$
4.0	$-0.866\imath$	$4.46 \cdot 10^{-16} - 0.929\imath$	$0.0117 + 2.09\imath$	$0.0116 + 2.09\imath$

Table 11: Case `1a` (cpv macos), $r = 4$, Z4: zeros.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00562 - 4.73 \cdot 10^{-4}\imath$	$0.00656 - 5.2 \cdot 10^{-4}\imath$
2.0	0	$5.68 \cdot 10^{-16} + 2.27 \cdot 10^{-16}\imath$	$0.00528 - 0.87\imath$	$0.00611 - 0.869\imath$
3.0	$0.866\imath$	$3.38 \cdot 10^{-18} - 0.929\imath$	$0.00792 + 0.871\imath$	$0.00875 + 0.87\imath$
4.0	$-0.866\imath$	$1.73 \cdot 10^{-16} + 0.929\imath$	$486.0 - 282.0\imath$	$539.0 - 319.0\imath$

Table 12: Case `1a` (cpv winos), $r = 4$, Z4: zeros.

3.2 Plots

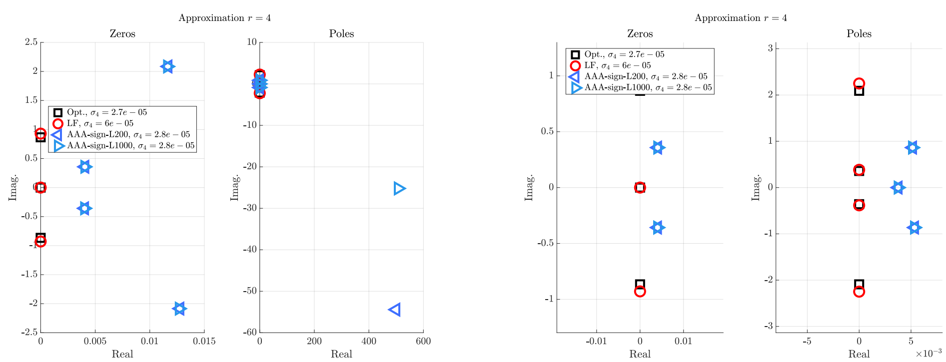

Figure 3: Case `'1a'`, $r = 4$. Left: zeros (left) and poles (right). Right: discarding the large values.

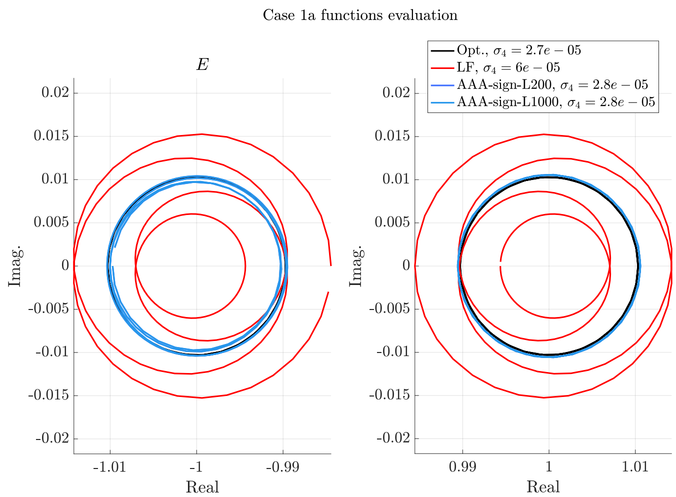

Figure 4: Case `'1a'`, $r = 4$. E and F evaluations.